

Jaehyung Jung

Postdoctoral researcher

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EXPERIENCES

- **Mobile Robotics Lab, Technical University of Munich** Sept. 2023 – Present
Postdoctoral Researcher (Supervisor: Prof. Stefan Leutenegger)
- **Air Force, Republic of Korea** Jan. 2011 – Jan 2013
F-15K Fighter Jet Maintenance Staff Sergeant

EDUCATION

- **Ph.D. Aerospace Engineering, Seoul National University** 2023
Thesis: Visual-Inertial Navigation System on Matrix Lie Group with Semantic Objects
- **M.S. Aerospace Engineering, Seoul National University** 2019
Thesis: Self-Calibrated Visual-Inertial Odometry for Rover Localization
- **B.S. Aerospace Engineering, Pusan National University** 2017
Magna Cum Laude (94%)

TECHNICAL SKILLS AND INTERESTS

Research interests: State estimation, Computer vision, Machine learning for robot perception

Programming: C++, ROS/ROS2, Python, Pytorch

Languages: English (C1), German (A1), Korean (native)

RESEARCH PROJECTS

- **AUTOASSESS** 2023 – Present
Scientific Assistant at TUM
 - Funded by Horizon Europe
 - Robot mapping with passive vision or LiDAR for vessel inspection
- **Indoor and outdoor integrated navigation technology** 2020 – 2023
Graduate Research Student at SNU
 - Funded by Korea Government
 - Uncertainty-aware visual-inertial simultaneous localization and mapping (SLAM)
 - Object pose estimation with shape symmetry
- **Pose estimation technology for mobile devices** 2018 – 2020
Graduate Research Student at SNU
 - Funded by Samsung Electronics
 - Event-inertial odometry for fast moving hand-held devices
- **Integrated navigation system for lunar rover** 2017 – 2018
Graduate Research Student at SNU
 - Funded by Korea Government
 - IMU / Camera fusion for planetary rover localization

OPENSOURCES

- **OKVIS2-X: Multi-sensor SLAM with volumetric dense mapping** – *IEEE Transactions on Robotics* 2025
- **Gaussian Mixture Midway-Merge for Object SLAM** – *IEEE Robotics and Automation Letters* 2022
- **Ensemble Visual-Inertial-Odometry** – *IEEE Transactions on Robotics* 2022

1. Simon Boche*, **Jae Hyung Jung***, Sebastián Barbas Laina*, and Stefan Leutenegger (*equal contribution), “OKVIS2-X: Open Keyframe-based Visual-Inertial SLAM Configurable with Dense Depth or LiDAR, and GNSS,” *IEEE Transactions on Robotics*, Accepted, 2025.
2. Min Seok Lee, **Jae Hyung Jung**, Ye Jun Kim, and Chan Gook Park, “Event-and Frame-based Visual-Inertial Odometry with Adaptive Filtering based on 8-DOF Warping Uncertainty,” *IEEE Robotics and Automation Letters*, vol. 9, no. 2, pp. 1003-1010, 2024.
3. **Jae Hyung Jung** and Chan Gook Park, “Gaussian Mixture Midway-Merge for Object SLAM with Pose Ambiguity,” *IEEE Robotics and Automation Letters*, vol. 8, no. 1, pp. 400-407, 2023
4. **Jae Hyung Jung**, Yeongkwon Choe, and Chan Gook Park, “Photometric Visual-Inertial Navigation with Uncertainty-Aware Ensembles,” *IEEE Transactions on Robotics*, vol. 38, no. 4, pp. 2039-2052, 2022.
5. **Jae Hyung Jung**, Jaehyuck Cha, Jae Young Chung, Tae Ihn Kim, Myung Hwan Seo, Sang Yeon Park, Jong Yun Yeo, and Chan Gook Park, “Monocular Visual-Inertial-Wheel Odometry using Low-Grade IMU in Urban Areas,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 2, pp. 925-938, 2022.
6. **Jae Hyung Jung**, Sejong Heo, and Chan Gook Park, “Observability Analysis of IMU Intrinsic Parameters in Stereo Visual-Inertial Odometry,” *IEEE Transactions on Instrumentation and Measurement*, vol. 69, no. 10, pp. 7530-7541, 2020.
7. **Jae Hyung Jung**, Sejong Heo, and Chan Gook Park, “Patch-based Stereo Direct Visual Odometry Robust to Illumination Changes,” *International Journal of Control, Automation, and Systems*, vol.17, no.3, pp. 743-751, 2019.
8. Sejong Heo, **Jae Hyung Jung**, and Chan Gook Park, “Consistent EKF-based visual-inertial navigation using points and lines,” *IEEE Sensors Journal*, vol.18, no.18, pp.7638-7649, 2018.

CONFERENCES & PREPRINTS

1. Sebastián Barbas Laina, Simon Boche, Sotiris Papatheodorou, Simon Schaefer, **Jaehyung Jung**, Stefan Leutenegger, “FindAnything: Open-Vocabulary and Object-Centric Mapping for Robot Exploration in Any Environment,” *arXiv*, 2025.
2. **Jae Hyung Jung**, Simon Boche, Sebastián Barbas Laina, and Stefan Leutenegger, “Uncertainty-Aware Visual-Inertial SLAM with Volumetric Occupancy Mapping,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
3. Hanyeol Lee, **Jae Hyung Jung**, and Chan Gook Park, “2D-3D Object Shape Alignment for Camera-Object Pose Compensation in Object-Visual SLAM,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
4. Min Seok Lee, Ye Jun Kim, **Jae Hyung Jung**, and Chan Gook Park, “Fusion of Events and Frames using 8-DOF Warping Model for Robust Feature Tracking,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
5. **Jae Hyung Jung**, and Chan Gook Park, “Object-based Visual-Inertial Navigation System on Matrix Lie Group,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2022.
6. Yeongkwon Choe, **Jae Hyung Jung**, and Chan Gook Park, "Ensemble Kalman Filter Based LiDAR Odometry for Skewed Point Clouds Using Scan Slicing," *IEEE International Conference on Robotics and Automation (ICRA)*, 2022.
7. **Jae Hyung Jung**, and Chan Gook Park, “Constrained Filtering-based Fusion of Images, Events, and Inertial Measurements for Pose Estimation,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2020.
8. **Jae Hyung Jung**, and Chan Gook Park, “Localization in High-Speed Motion Using IMU-aided Event Flow Estimation,” *The Institute of Navigation GNSS+ (ION GNSS+)*, 2020.

ACADEMIC ACTIVITIES

- **Co-organizing Workshop in ICRA 2025** 2025
Novel Approaches for Precision Agriculture and Forestry with Autonomous Robots
- **Organizing Workshop in RSS 2024** 2024
Navigation & Mobile Manipulation in Challenging and Cluttered Natural Environments
- **A reviewer in ICRA, IROS, RSS, RA-L, T-RO** 2021 – Present

AWARDS

- **Best Paper Award** in Avionics Systems Symposium Korea *2019*
- **Best Paper Award** in 33rd Institute of Control, Robotics, and Systems, Annual Conference *2018*
- **The 2nd Prize** in CANSAT competition Korea *2016*
 - Organized by Ministry of Science and ICT
 - Team “To the Space!”: 2-DOF camera gimbal stabilizer for CANSAT